

Endogenous Security Techniques for Safeguarding Next-Generation Spectrum Usage at the Physical Layer

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ABSTRACT

The rapid increase in the data transmission rates for next-generation communication systems has intensified the demand for spectrum, straining limited resources under traditional management. This has spurred a move toward dynamic spectrum sharing and exploration of higher frequency bands. Although spectrum sharing enables other users to access the underutilized whitespace spectrum, it also raises additional security concerns. The possibility of attacks at higher carrier frequencies has not yet been well characterized. To address these challenges, endogenous properties of wireless communication systems can be explored as new opportunities for security design. The purpose of this tutorial is to provide a comprehensive overview of the state-of-the-art wireless endogenous security techniques that guarantee wireless security and privacy for spectrum usage at the physical layer, along with the principles, solutions, and future research trends.

INTRODUCTION

The upcoming mobile wireless era faces spectrum and bandwidth constraints due to growing demands for lower latency, better mobility, dense connectivity, and higher data rates. This scarcity has led researchers to thoroughly investigate spectrum sharing techniques. These techniques enable simultaneous use by multiple users without interference, thereby improving the efficiency and flexibility of spectrum utilization and management [1]. Moreover, the wireless communication industry is actively exploring the use of unlicensed spectrum at elevated carrier frequencies, such as the terahertz (THz) range exceeding 100 gigahertz [2], to address the scarcity of spectrum and accommodate the increasing demands of data traffic.

Although the aforementioned spectrum usage methods offer benefits, sharing or newly accessed spectrum may involve various potential risks. For example, in the process of dynamic spectrum access (DSA), attackers can deceive

second users (SUs), forcing them to misuse non-idle bands, interfere with primary services, or force other SUs to leave bands. Despite the significant propagation loss of the high-frequency electromagnetic wave, which could be leveraged for inherently secure communication channels, the issue of surface scattering becomes pronounced and could result in critical signal eavesdropping. Given the critical role of communication devices in daily life and their storage of vital information, it is essential to implement strong security and privacy protocols to protect data, resources, and users from threats and unauthorized spectrum access.

Existing spectrum security defense methods, relying heavily on external encryption and protocol protection, are reactive and layered. They can hardly counter dynamic and complex attacks and struggle to optimize security and communication performance together. For example, in current spectrum usage, a device's identity is not strongly associated with its intrinsic properties. Consequently, an adversary can avoid detection even when initially caught by altering their identity and staging a subsequent attack, thereby undermining defense capabilities.

Fortunately, the communication system has recently been discovered to possess endogenous secure attributes. The endogenous security theory developed from these attributes can be employed not only to mitigate known security threats but, more importantly, to counteract unknown attacks [3]. To this end, some endogenous security-based solutions are proposed to strengthen security at the network layer. These solutions not only ensure the reliability of the system, but also obfuscate the structure and operational mechanisms of the system from potential attackers, thereby complicating the execution of effective attacks.

Driven by this motivation, this paper focuses on the endogenous security characteristics of wireless communication systems, with the aim of maximizing their positive impact on spectrum utilization protection. Although earlier studies have explored the idea of endogenous wireless security

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[3], their attention is limited to the wireless channel alone. Given the uncertainties, instability, and unpredictable security threats, greater emphasis is urgently required on developing endogenous attributes to address a range of spectrum threats.

The contributions of this article are three-fold:

- We conducted a comprehensive evaluation of security vulnerabilities in both existing low-frequency spectrum resources and emerging high-frequency bands suited for next-generation wireless communication systems.
- We have devised a novel and comprehensive endogenous security framework at the physical layer, thoroughly examining how such features enhance wireless communication security.
- Using the endogenous security framework, we designed solutions to tackle spectrum data leaks, unauthorized spectrum use, and interference issues. This approach establishes a comprehensive framework for secure spectrum use, improving system resilience in complex defense scenarios.

ADVANCED SPECTRUM UTILIZATION TECHNOLOGIES: ASSESSMENT OF SECURITY VULNERABILITIES

This chapter introduces technologies crucial for optimizing spectrum usage to meet the needs of future wireless communications while mitigating related security risks.

ADVANCED TECHNOLOGIES FOR SPECTRUM USAGE

It is essential to develop advanced strategies to enhance the efficiency of existing spectrum resources while also investigating higher frequency bands and expanding the spectrum available.

Current Spectrum Resources

Traditional static spectrum allocation and orthogonal multiple access methods are now seen as inefficient, as they often waste resources by giving too much spectrum to a single user who may not fully use it. Therefore, the following technologies are used to improve spectrum efficiency.

a) Dynamic Spectrum Access and Sharing:

For spectrum allocation, DSA is crucial to improve efficiency by letting unlicensed users access unused licensed spectrum, enhancing utilization. In CRNs (Cognitive Radio Networks), licensed spectrum holders are PUs (Primary Users), while unlicensed users who can access others' spectrum without disruption are SUs (Secondary Users). Various DSS (Dynamic Spectrum Sharing) schemes, like LSA (Licensed Shared Access) and CBRS (Citizen Broadband Radio Service), have been explored within CRNs.

b) Non-Orthogonal Multiple Access: Beyond DSA, advanced multiple-access techniques like NOMA (Non-Orthogonal Multiple Access) can further boost spectrum utilization. NOMA uses non-orthogonal resource allocation and signal superposition in the same wireless resources, relying on power domain multiplexing. Reception is achieved via SIC (Serial Interference Cancellation) for signal detection and demultiplexing, making NOMA effective for increasing user capacity in future wireless systems.

High-Frequency Spectrum Bands

The crowded nature of sub- 6GHz frequencies requires investigation of alternative frequency bands to meet emerging requirements [4].

a) Millimeter Wave: The millimeter wave (mmWave) band, spanning 30 to 300 GHz with wavelengths from 1 to 10 millimeters, offers a bandwidth significantly exceeding that of sub-6GHz bands. Due to the logarithmic relationship between bandwidth and data rates in the Shannon-Hartley theorem, mmWave communication can achieve data rates far higher than current standards [5]. This makes mmWave attractive for next-generation wireless communications and applications like aerial and vehicular communications.

b) Terahertz: THz communication, spanning the extensive frequency range of 0.1 to 10 THz, is becoming a crucial technology for 6G and future advancements. The IEEE 802.15.3d standard (approved in 2017) enables 100 Gbit/s data rates for switchable point-to-point links around 300 GHz [6]. THz's vast spectral capacity boosts high-throughput applications, and its short wavelength supports micro- and nano-scale communication, enhancing wireless precision.

c) Optical Wireless Communication: Optical frequency spectrums can advance high-density, high-capacity 5G and Internet of Things (IoT) networks. OWC (Optical Wireless Communication) uses these spectrums for high-speed data, increased capacity, and reliability. Its wide frequency range, fast transmission, low latency, strong security, cost-effectiveness, and energy savings make it ideal for future wireless needs. The extensive light wavelength spectrum supports diverse communication models, like device-to-device and vehicle-to-vehicle links.

SECURITY ISSUES IN SPECTRUM USAGE

Although these approaches address spectrum scarcity, they face distinct security vulnerabilities for current low-frequency spectrum resources and the arising high-frequency spectrum bands, as illustrated in Fig. 1.

Spectral Data Leakage

The widespread issue of spectrum data leakage, arising from eavesdropping and various techniques, presents itself differently in low-frequency versus high-frequency communications.

a) Eavesdropping: Inevitably, spectrum sharing raises the susceptibility to eavesdropping attacks, which are generally divided into active and passive eavesdropping. When the transmitter emits signals in all directions without distortion, passive eavesdroppers with advanced receivers could potentially intercept the signals. For example, in NOMA settings, the superposition of signals facilitates increased interception due to simultaneous transmissions. Active eavesdroppers, in contrast to passive eavesdroppers, not only intercept sensitive spectrum data but also intentionally cause disruptions. They interfere with communications by transmitting disruptive signals or disseminating fraudulent pilot sequences. In addition, they undermine channel secrecy and potentially reduce channel secrecy capacity to zero.

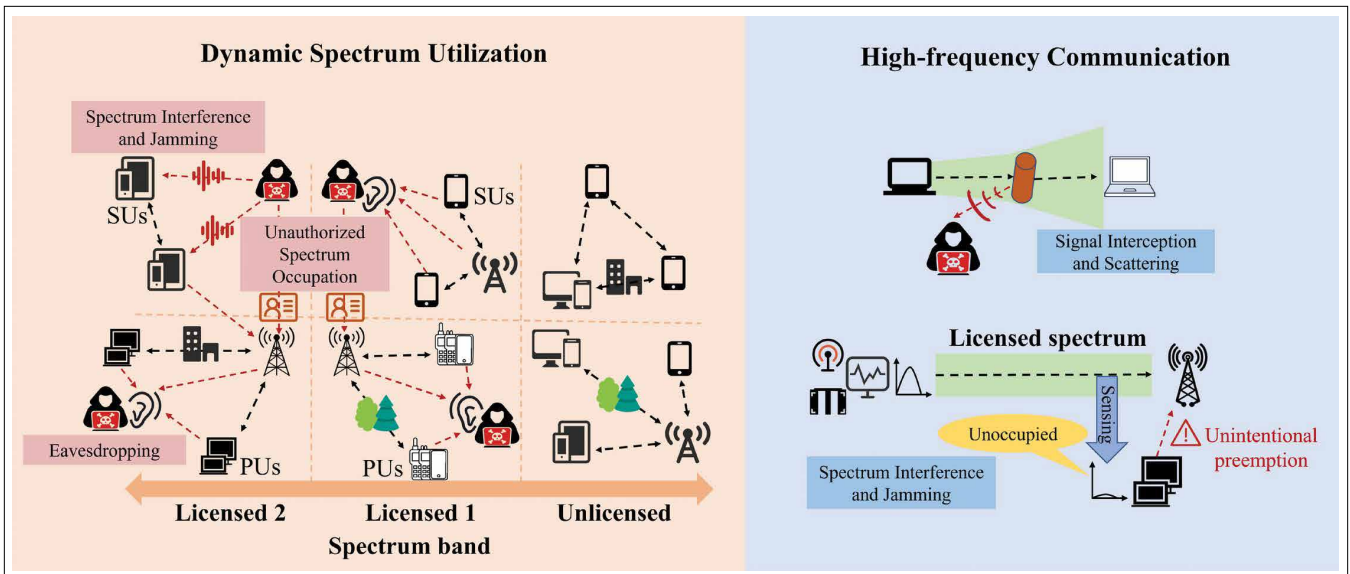


FIGURE 1. Security challenges in future spectrum usage include low-frequency communication issues such as spectral data leakage, unauthorized spectrum occupation, and spectrum interference and jamming, all pertaining to the dynamic spectrum utilization of existing spectrum resources. In contrast, high-frequency communication mainly addresses concerns about data leakage caused by signal scattering and spectrum interference.

b) Signal Interception and Scattering: Unlike low-frequency eavesdropping, where signals are intercepted differently, high-frequency eavesdroppers are able to intercept transmissions by utilizing small objects to passively scatter signals along the line-of-sight (LOS) trajectory. In non-line-of-sight (NLOS) conditions, THz waves are prone to scattering when they encounter surface inconsistencies, which results in their dispersion across multiple directions. For instance, adversaries can take advantage of Bragg's diffraction law to alter the direction of beams through the strategic use of reflectors, thus making high-frequency communication systems vulnerable to interception and signal capture [7].

Spectrum Illegal Occupation

Once attackers acquire sensitive information, they are quite likely to misuse the data from real users to unlawfully capture dynamically allocated spectrum.

a) Spectrum Hijacking and Forgery: Self-interested parties can exploit spectrum sharing by posing as authorized users prior to the initiation of security measures, allowing unauthorized occupation of spectrum in standalone networks. In CR contexts, attackers manipulate the sensing data by introducing fake reports. This false information can sway secondary users or the fusion center, consequently affecting decisions regarding frequency bands. Furthermore, because of the open nature of DSS, legislators with self-serving motives might illegally use spectrum beyond permitted limits for personal benefit.

b) User Emulation Attack: In cognitive radio (CR) systems, user emulation attacks, where adversaries mimic legitimate user characteristics to send signals with similar features, are especially common as primary user emulation attacks. These attacks typically target the spectrum sensing phase, with SUs mimicking PUs to transmit on licensed bands. The primary network's security

mechanisms lack the capability to authenticate primary user emulation access, resulting in missed spectrum opportunities. The user emulation attack also allows the initiation of sophisticated attacks, such as spectrum honeypots.

Spectrum Interference and Jamming

In addition to occupying the spectrum, attackers can disrupt or completely block normal communication. The following section details the security risks related to such disruptions.

a) Harmful Jamming Signal Insertion: The introduction of interference signals poses a security threat in dynamic spectrum utilization. Within NOMA frameworks, where interference is inherently limited, adversaries can introduce disruptive signals that decrease the data rates of legitimate users and increase eavesdropping risks. In CR scenarios, interference jeopardizes both spectrum sensing and selection. In mmWave settings, jammers can conduct denial of service (DoS) attacks by broadcasting on the receiver's frequency, leading to channel congestion.

b) Interference Caused by Attenuation: Significant attenuation of the primary user's signal, notably when it matches the secondary user's power level, leads to interference. Multipath fading exacerbates this issue, undermining spectrum sensing reliability and increasing incorrect spectrum access risks. Higher frequency bands worsen the problem due to greater path and penetration losses [8].

These security issues constitute major barriers to spectrum usage, highlighting the pressing necessity for effective mitigation approaches to safeguard radio frequency resources access.

ENDOGENOUS SECURITY ATTRIBUTES OF WIRELESS COMMUNICATION SYSTEMS

This chapter investigates the endogenous security attributes of wireless communication systems to

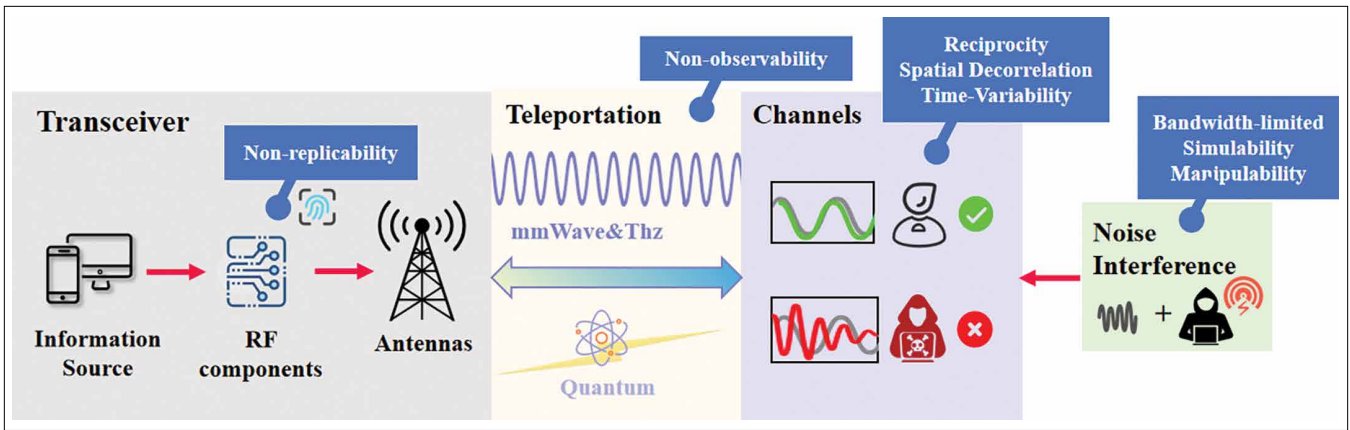


FIGURE 2. The endogenous security attributes of a typical wireless communication system.

address these threats, categorizing their components into transceivers, teleportation, channels, and noise/interference. Fig. 2 depicts these endogenous security attributes of a standard wireless communication system framework.

TRANSCIVER

In wireless communication, transceivers are the hardware crucial for sending and receiving signals. The endogenous security attributes of transceivers comprise:

- **Uniqueness:** Differences in manufacturing and drift tolerances lead transceivers to have inherent physical flaws, making each device uniquely non-ideal. This enables device identity to be uniquely identified at the physical layer, with such identity naturally generated and independent of external factors.
- **Robustness:** The physical characteristics of a device remain stable over an extended period, making them suitable for use as authentication credentials. These characteristics can be recorded in a database and then extracted for authentication when needed.
- **Non-Replicability:** The intrinsic properties of these devices are challenging to accurately replicate or mimic. While attackers may intercept signals, reproducing a device's physical characteristics is nearly impossible due to receiver uncertainty, thereby preventing identity theft.

Emerging techniques like radio frequency fingerprint (RFF) and physical unclonable functions (PUF) possess potential as identity authentication tools due to these features. Their main difference lies in the fact that RFF is a passive identification technology that does not require hardware modification, while PUF is an active security technology that requires the integration of specific circuit structures during the chip design phase. Since RFF does not require the embedding of dedicated hardware modules, its deployment cost is relatively low. PUF systems generally require specific protocol procedures, such as challenge-response protocols, to verify the device's identity. PUF can also generate unique keys for chips, which can be used for secure needs such as encrypted communication and anti-counterfeiting.

TELEPORTATION

With technological advancements, communication has evolved from traditional electromagnetic wave communication to higher-frequency mmWave and THz bands, and has even entered the era of quantum communication.

The characteristic of non-observability serves as an endogenous security mechanism within the realm of electromagnetic wave teleportation, particularly within the domains of high-frequency bands. In such bands, significant signal attenuation coupled with the rapid fading phenomenon substantially diminishes the likelihood of signal interception, thereby enhancing overall communication security.

Quantum communication further enhances non-observability by leveraging unique quantum properties. This phenomenon can be deconstructed through the following aspects:

- **Non-Clonability:** Quantum states cannot be cloned or reproduced, preventing attackers from duplicating transmitted information and ensuring confidentiality.
- **Superposition:** Quantum superposition allows particles to exist in multiple states until measured. Any observation collapses the state and alters the information, ensuring that unauthorized access inherently disrupts transmission and guarantees security.
- **Entanglement:** Entangled particles are instantaneously correlated, allowing state transfer without direct transmission, thus enhancing the security and reliability of communication.

CHANNELS

In wireless networks, the channel plays a vital role in data transmission, underpinning both system design and security. The inherent reciprocity, spatial decorrelation, and time-variance characteristics of wireless channels offer intrinsic security advantages for communication systems, independent of external security protocols or intricate encryption techniques.

- **Reciprocity:** Channel reciprocity refers to the symmetry between transmission and reception paths for communicating parties, enabling legitimate users to create shared keys using channel data while complicating synchronized eavesdropping by attackers.

- **Spatial Decorrelation:** Channel spatial decorrelation introduces notable discrepancies in channel state information (CSI) between attackers at distinct positions and legitimate users, significantly complicating eavesdropping efforts.
- **Time-Variability:** Channel time-variability provides legitimate users with high entropy random sources, improving the randomness and security of key generation while simultaneously increasing the difficulty of attackers in maintaining consistent comprehension of the channel state.

Due to the inherent characteristics of wireless channels, wireless communication systems enable two intrinsic security strategies: channel fingerprint authentication and shared key generation based on channel reciprocity. The first method leverages the spatial decorrelation of the channel, allowing near-instantaneous security confirmation through real-time CSI measurement and comparison. The latter avoids the dependence on complex cryptographic protocols and trusted third parties in traditional key distribution and provides natural anti-eavesdropping and antiforgery capabilities.

INTERFERENCE AND NOISE

In the communication process, noise and interference are inevitable physical phenomena that typically degrade signal quality and affect the performance of the communication system. However, by leveraging the characteristics of different types of interference and noise, various technical approaches can be used to effectively defend against and protect the security of communication systems.

With regard to interference, spread spectrum techniques utilize the bandwidth-limited nature of noise by distributing signals across broader spectral ranges, extending signals across multiple frequency bands, and reducing interference concentration zones. This characteristic of spectral dispersion makes it difficult for attackers to effectively interfere with signals at specific frequency segments, thereby improving resistance to interference from the communication system.

When it comes to noise, three intrinsic characteristics support protective measures for maintaining spectrum security.

- **Suppressibility:** Beamforming techniques can direct noise in the directions of eavesdroppers, maintaining high signal-to-noise ratios in legitimate communication channels while significantly attenuating the quality of the eavesdropping channel.
- **Artificial Augmentability:** Interference-assisted techniques artificially inject noise into specific frequency segments or spatial directions, making signal acquisition and stable decoding difficult for eavesdroppers.
- **Simulability :** Noise simulability lays the theoretical foundation for covert communication, allowing advanced signal designs to conceal information within noise, thereby boosting system security.

Building on these endogenous properties of the wireless system, we have designed a protection framework that strategically applies these features to tackle security issues at various communication stages. This results in a layered

defense that significantly increases the security and resilience of the system.

ENDOGENOUS SECURITY SOLUTIONS FOR SAFEGUARDING SPECTRUM UTILIZATION AT THE PHYSICAL LAYER

As shown in Fig. 3, the endogenous security attributes discussed in Chapter III can be exploited to address spectrum vulnerabilities.

SPECTRUM DATA LEAKAGE

The core of spectrum data leakage lies in attackers attempting to obtain communication content through passive listening or active intervention, thus compromising communication confidentiality. For this spectrum security threat, rational utilization of endogenous wireless security technologies can provide effective protection.

Secret Key Generation

To prevent the possibility of spectrum data leakage, we utilize endogenous wireless security features associated with both teleportation and channels. These attributes are employed to produce identical and unique keys. These keys are then used for encrypting the spectrum data, thus ensuring robust protection against any leakage of spectrum-related data.

Due to the unobservable nature of quantum states, quantum key distribution (QKD) methods are well-suited for secure communications. The attempted eavesdropping by attackers would be identified, guaranteeing a secure key distribution, even against powerful computational adversaries. QKD infrastructure has been increasingly integrated into backbone networks; however, its implementation in IoT networks' last mile remains challenging.

Channel reciprocity-based key generation (CRKG) offers a lightweight and easily distributed security solution for large-scale IoT access settings. Due to the channel's time-variance and spatial decorrelation, keys are dynamically updated and possess high spatial diversity, rendering eavesdropping challenging to predict or reconstruct. Although this technique may face challenges in environments with static and wave blockage conditions, the new reconfigurable intelligent surface (RIS) technology shows potential to improve CRKG by shaping the wireless environment [9]. Fig. 4(a) illustrates the rate of secure key generation in scenarios where many users are distributed in a dense manner. The findings demonstrate that secure data transmission is effectively achieved, even when dealing with channel correlation, while preventing any leakage across the spectrum.

The authors in [10] presented channel reciprocity-based quantum key distribution (CR-QKD), enabling secret key sharing over wide-area mobile networks by combining these two techniques.

Signal Quality Defense

Previous key generation technology guarantees the impossibility of decryption even when eavesdroppers obtain a signal quality sufficient for decryption, while signal quality defense mechanisms control quality to render intercepted signals insufficiently decodable.

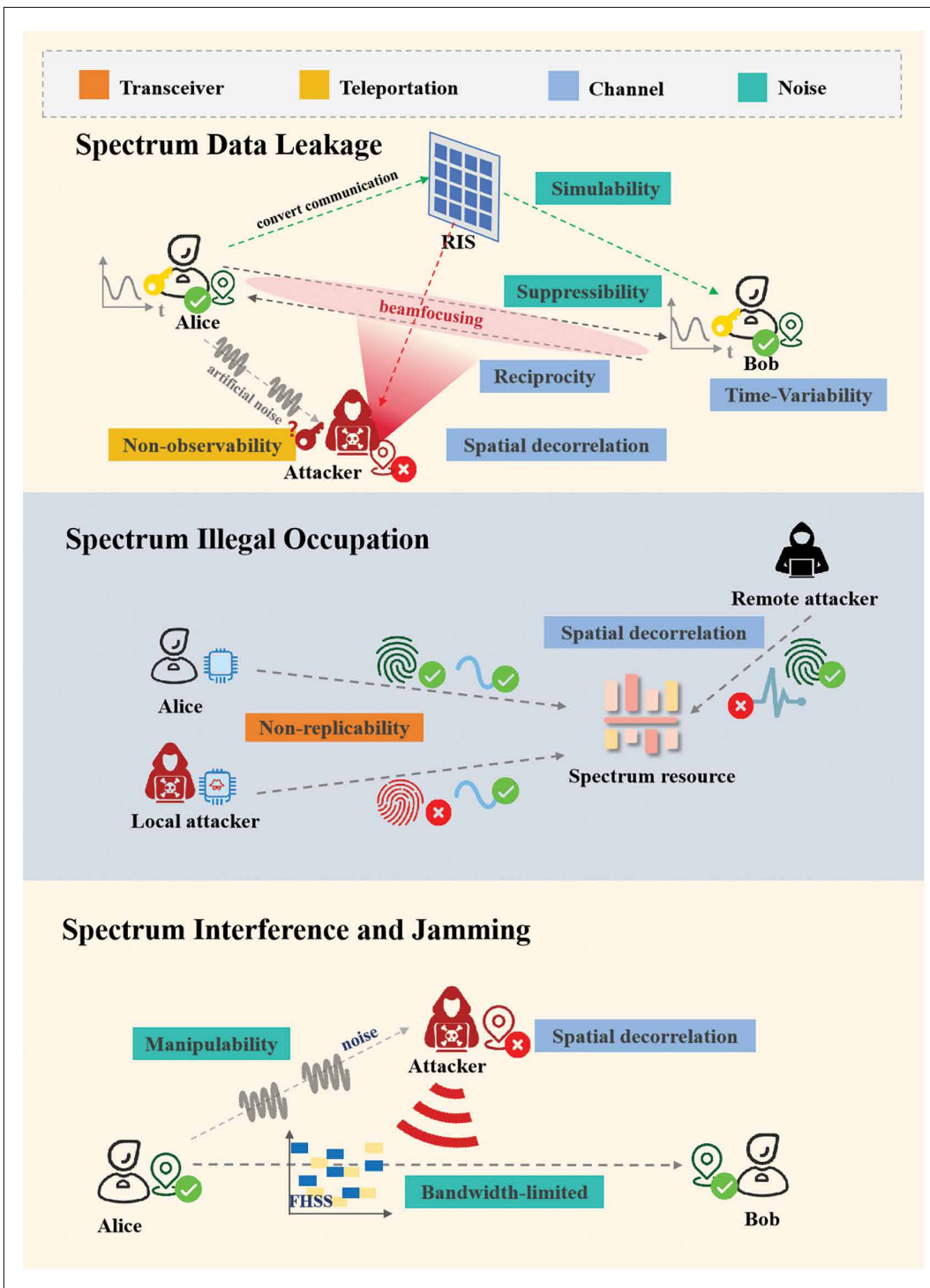


FIGURE 3. Endogenous Security Solutions for Safeguarding Spectrum Utilization at the Physical Layer.

In short-range or privacy-sensitive settings, mmWave and THz technologies enhance spectrum protection through transmission unobservability. These ultrahigh-frequency communications have highly directional signals and fast fading, challenging attackers' ability to reliably intercept or manipulate the signal, thus restricting eavesdropping, hijacking, or forgery attempts.

In addition to the inherent benefits of high-frequency communication, external help can employ noise suppression for beamforming. As 6G communication needs grow, conventional

far-field beam control, which relies on plane wave assumptions, is inadequate for precise signal focusing and interference mitigation. In near-field zones (Fresnel zones), electromagnetic waves have spherical wavefronts, allowing transmitters to direct signal energy to defined angles and depths. This focusing feature offers an additional means to isolate interference by angle and range. Beam focusing achieves directional signal energy transmission through spatial precoding of transmitted signals. By enhancing signal power in legitimate user directions while establishing power

suppression regions in potential interference directions, it can significantly reduce the impact of interference on communication systems.

Covert Communication

Although the two protection methods above can prevent eavesdroppers from decrypting or decoding information, they may not stop signal interception during transmission. To enhance the security of the communication process, covert communication offers a novel approach, particularly suitable for military communications or privacy-sensitive data transmission scenarios.

Covert communication considers a communication model comprising Alice (transmitter), Bob (receiver), and Willie (monitor), where Alice aims to transmit information to Bob without being detected by Willie. Therefore, its objective is to conceal the transmission behavior itself, rather than simply encrypting the transmission content, thus providing a higher level of protection [11]. By adjusting signal power control, introducing artificial noise (AN), or utilizing RIS auxiliary technologies, communication behavior characteristics can be effectively masked, making it challenging for attackers to perceive the communication's existence.

SPECTRUM ILLEGAL OCCUPATION

Unauthorized spectrum occupation seeks to undermine the integrity and trust of wireless communications by unlawfully occupying spectrum or masquerading as legitimate devices. Attackers could exploit weaknesses in spectrum resource management to hijack or forge signals, misleading legitimate communication identification systems. This threat is particularly significant in DSS high-frequency communication scenarios.

Hardware Feature Approach

We examine the RFF and PUF methods that leverage the physical properties of wireless communication devices for authentication.

RFF effectively combats illegal spectrum occupation through unique physical-layer features. As

illustrated in Fig. 4(b) [12], models that utilize the RFF approach consistently perform well across different sample sizes in actual ADS-B signal environments. Using sophisticated small-sample learning techniques, they achieve detection accuracies that exceed 90% even with only 30 labeled samples per device. Concurrently, basic CNN models continue to exhibit strong performance levels when presented with ample labeled data. These findings underscore the versatility and resilience of RFF in both limited and extensive data conditions, positioning it as a fundamental tool in tackling unauthorized spectrum usage.

PUF plays a significant role in hardware security, optimizing key management, data encryption, and identity authentication. RFF and PUF together enable layered identity verification, improved key management, and stronger attack resistance. Based on related research, Zhang et al. [13] proposed a collaborative mutual authentication and key establishment protocol combining PUF and RFF, using RFF to confirm the origin of the transmission and using PUF for secret exchange and authentication at device level, effectively reducing communication overhead and strengthening resistance to man-in-the-middle and replay attacks.

Channel Feature Approach

Currently, the reliability of device identification based on hardware features like RFF and PUF is insufficient. With an increasing number of users, the likelihood of device characteristic overlap and collision grows. By leveraging channel feature identification, this issue is mitigated since variations in device channel CSI by location enable the detection of remote attacks. Existing channel fingerprinting technologies primarily categorize into passive and active protection methods [14].

a) Passive Protection Scheme: In DSS networks, spectrum protection is ensured by employing CSI for spectrum access criteria. This involves evaluating geographic locations and channel properties (e.g., multipath or fading) of

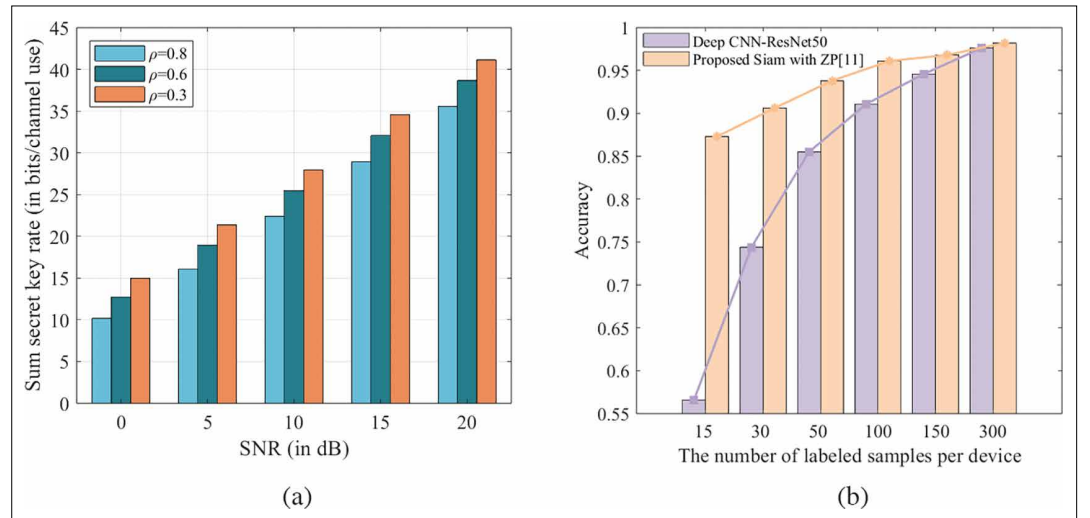


FIGURE 4. Defense strategies against Spectrum Leakage and Illegal Spectrum Occupation. a) Sum secret key generation rate under different user's channel correlation ρ in dense user deployment scenarios without spectrum leakage. b) Classification accuracy across ten categories of real-world ADS-B signals, with different numbers of the labeled samples per device under different models.

the parties to verify device authenticity and block unauthorized devices. The channel characteristics' time-varying, spatially decorrelated nature makes it difficult for attackers to mimic a legitimate device's channel fingerprint, effectively thwarting spectrum hijacking and forgery.

b) Active Protection Scheme: Unlike passive methods, active strategies embed authentication tags within signals using shared key generation. Receivers verify the legitimacy of communication counterparts by comparing these tags with pre-defined key-matching criteria, thereby blocking forged signals and unauthorized device access. In environments where spectrum sharing or competition for resources occurs, active authentication quickly detects falsified signals, improving communication system security.

SPECTRUM INTERFERENCE AND JAMMING

Active Interference Defense

Frequency Hopping Spread Spectrum (FHSS) utilizes the limitations of noise bandwidth by having communication parties constantly change carrier frequencies according to predetermined hopping sequences to minimize interference effects. Yet, the conventional FHSS approach loses effectiveness when interference sources are capable of monitoring signal hopping patterns. Shi et al. [15] introduced the Index Modulation Frequency Hopping Spread Spectrum (IM-FHSS) technique, integrating index modulation with conventional FHSS by altering signal activation states to improve interference resistance. IM-FHSS combines index modulation with conventional FHSS, improving interference resistance by changing signal activation states. In reactive interference cases, IM-FHSS masks idle frequencies, complicating the interference source's tracking and attack.

Under constant interference, it uses interference signals as auxiliary energy by tweaking transmission power. For random interference, IM-FHSS opts for minimal available frequencies, enhancing interference resistance while conserving spectrum resources. The experimental results in [15] shows that IM-FHSS has superior robustness compared to traditional FHSS across different interference conditions, highlighting the effectiveness of this active defense strategy.

Passive Interference Defense

Passive interference defense mechanism integrates channel fingerprinting with beamforming techniques to reduce interference. Wireless channels' spatial decorrelation provides unique CSI for signals from the same device at different locations. This allows extracting key characteristics like path loss, shadowing, and small-scale fading as distinct channel fingerprints for device localization. Even if an attacker uses advanced techniques like generative adversarial networks (GANs) to simulate legitimate user identities, the inherent CSI differences due to physical separation can reveal their true location, ensuring strong attack traceability. Once the attacker's location is determined, the controllability of noise can be exploited for targeted beamforming, directing interference suppression precisely toward the attacker's direction. This coordinated approach-first utilizing CSI for attacker localization and then employing beamforming to mitigate interference-forms a robust passive defense against jamming attacks.

Table 1 provides a comprehensive overview of these endogenous security solutions designed to safeguard spectrum usage at the physical layer for the aforementioned security issues including spectrum data leakage, spectrum illegal occupation and spectrum interference and jamming. It

Security Issues	Solution Technology	Endogenous Security Attributes	Advantages & Disadvantages	Possible Solutions
Spectrum data leakage	Secret key generation	- Channel: reciprocity, spatial decorrelation, and time-variability - Teleportation: transmission state non-observability	✓ Dynamically updated and unique keys ✓ Infeasible decryption even with eavesdropping ✗ Require channels that satisfy reciprocity	AI channel enhancement techniques to mitigate non-reciprocity
	Signal quality defense	- Noise: suppressibility - Teleportation: transmission state non-observability	✓ Suppress the power of eavesdropping signals ✓ Insufficient decoding of intercepted signals ✗ Signal attenuation due to high-frequency communication	RIS-assisted reflection and HMIMO to enhance propagation
	Convert communication	Noise: simulability	✓ Render the communication process invisible to attackers ✗ Additional resource expenditure	Net zero communication to reduce total power consumption.
Spectrum illegal occupation	Hardware feature approach	Transceiver: non-replicability	✓ Unclonable physical characteristics ✓ Reduce communication overhead while enhancing security ✗ RFF may experience degradation due to the accumulation of usage time	Update RFFs timely to maintain recognition accuracy.
	Channel feature approach	Channel: spatial decorrelation	✓ Achieve the detection of remote attacks ✓ Complement RFF authentication ✗ Challenging to be employed independently	RF-channel fingerprint fusion to enhance reliability.
Spectrum interference and jamming	Active interference defense	Noise: bandwidth-limited nature	✓ Flexible countermeasures of IM-FHSS ✗ Dependence on Cognitive Disruption	Distributed detection across multiple nodes to improve robustness.
	Passive interference defense	- Noise: manipulability - Channel: spatial decorrelation	✓ Attenuate eavesdropping channel quality ✓ Suitable for dynamic eavesdropping environments ✓ Acquire CSI revealing geographic location data ✗ Dependence on precise CSI	Leverage azimuth angle of arrival (AAoA) and elevation angle of arrival (EAoA) to reduce reliance on CSI.

TABLE 1. Summary of endogenous security solutions for safeguarding spectrum utilization at the physical layer. Green correct marks and red cross marks present advantages and disadvantages respectively.

elaborates on the endogenous security attributes, advantages, and disadvantages associated with these mechanisms. Additionally, the table suggests potential solutions to mitigate the identified drawbacks. In prospective developments, these solutions are expected to evolve into an extensive endogenous security protection framework,

which is capable of efficiently mitigating particular threats while providing substantial security in intricate and varied spectrum usage contexts.

CONCLUSION AND FUTURE DIRECTIONS

This paper thoroughly evaluated security vulnerabilities, including data leaks, unauthorized access,

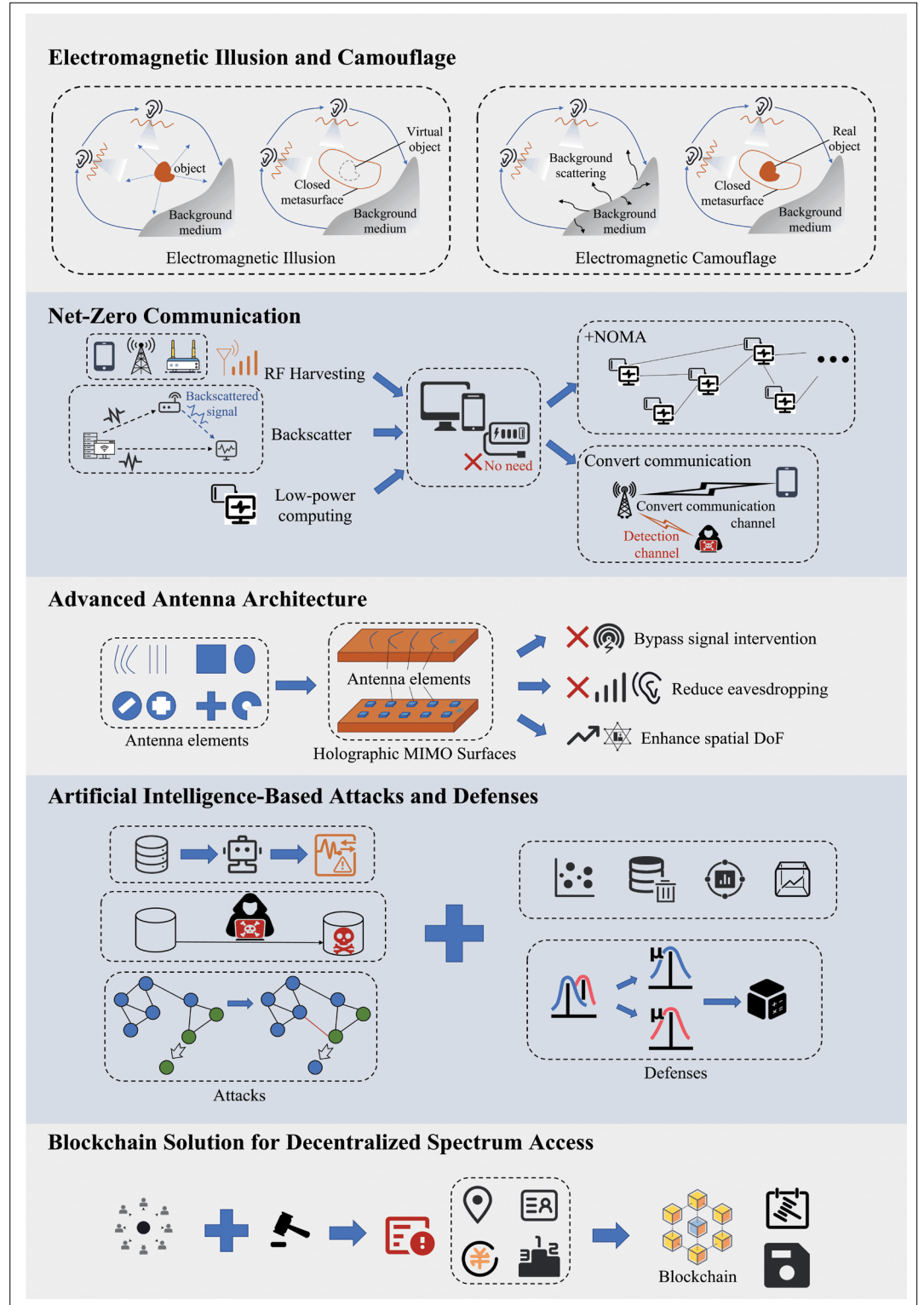


FIGURE 5. Future directions: prospective technological trajectories and research avenues, mainly including electromagnetic illusion and camouflage, net zero communication, advanced antenna architecture, AI-based attacks and defenses, and blockchain solution for decentralized spectrum access.

and disruptions in current low- and high-frequency bands for next-generation wireless systems. To address these problems, this article proposed an innovative endogenous security framework at the physical layer, analyzing the security attributes of wireless communication systems thoroughly by categorizing their components into transceivers, teleportation, channels, and noise/interference. Using these endogenous security attributes, this article proposed solutions to address security issues in next-generation spectrum usage, leading to a physical-layer spectrum framework grounded in endogenous security.

Fig. 5 describes the potential research directions for future study.

ELECTROMAGNETIC ILLUSION AND CAMOUFLAGE

Electromagnetic illusion and camouflage are innovative security techniques that not only conceal communication devices and signals, but also actively mislead adversaries, offering superior security. Illusion technology creates virtual objects or alters forms to prevent unauthorized spectrum detection, enhancing protection and disguise. By generating misleading communication signals and creating false communication targets, it exploits the inherent security properties to mislead sensors and antennas, making it difficult for adversaries to determine the true source of the signal. Multi-spectral adaptability is crucial for achieving electromagnetic camouflage and illusion, yet it poses challenges due to differing principles across bands and the necessity for comprehensive coverage. Currently, no materials meet these demands, indicating the need for further technological development.

NET ZERO COMMUNICATION

Net-zero communication significantly enhances covert communication using an innovative endogenous security approach. This relies on technologies such as radio frequency energy harvesting, backscatter, and low power computing to operate wireless devices without rechargeable power. Advancing this is vital because integrating it with NOMA can increase connectivity for low-power devices in future networks. In addition, its low power nature is ideal for secure communications by hiding signals below the noise floor, using noise for security to thwart eavesdropping. However, challenges include limited computational and transmission capacities, which complicate the implementation of complex security protocols.

ADVANCED ANTENNA ARCHITECTURE

Holographic multiple input multiple output (HMIMO) is a cutting-edge antenna design that can enhance spectrum security by minimizing eavesdropping and blocking spectral interference. It is enabled by metamaterials and metasurfaces that employ holography-based, ultrathin, large-scale, continuous leaky-wave antennas or photonic tightly coupled arrays with sub-wavelength components. With a continuous spatial aperture, HMIMO processes electromagnetic signals directly, eliminating digital interference. This approach minimizes wave reflections and refractions, thus improving security by reducing signals perceivable by eavesdroppers. Additionally, as

spectrum demand grows, high-frequency signals face severe attenuation and detection challenges. HMIMO's dense antenna arrays and beamforming enhance propagation, improving coverage and security in high-frequency communications.

ARTIFICIAL INTELLIGENCE-BASED ATTACKS AND DEFENSES

AI is widely utilized in endogenous security technologies, but it also poses new challenges for future security. The broad use of AI in data processing heightens privacy issues caused by unclear processes, amplifying data breach risks. Furthermore, its security weaknesses make it vulnerable to training-phase poisoning and inference-phase adversaries. Lastly, AI's advanced search capabilities enhance its ability to discover security flaws and execute targeted attacks.

To counter AI-based attacks, defensive tactics involve disrupting corrupted data characteristics via data clustering, cleaning, augmentation, and strengthening model robustness, thus diminishing their impact. For adversarial examples, the analysis of the differences in the probability distribution helps in the training of resilient models against such attacks.

BLOCKCHAIN SOLUTION FOR DECENTRALIZED SPECTRUM ACCESS

In decentralized DSS, spectrum auctions are crucial for reallocating and dynamically accessing spectrum. Nevertheless, privacy issues are often ignored, with security threats targeting sensitive details like bids, identities, rankings, locations, and timings, along with risks of spectrum tampering. Blockchain acts as a secure ledger, ensuring immutable and verifiable records of authorized auctions, thereby protecting all auction data. By fostering decentralized networks among untrusting entities, blockchain provides an automated platform for spectrum allocation and regulation, making it a promising area for research in combining endogenous security with blockchain technology.

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